

SIMATS ENGINEERING

C03-ASSESSMENT 2

Port Scanning Challenge

ITA1407-ETHICAL HACKING

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COURSE OUTCOME COVERED

CO3: Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BT4)

Assessment Tool Weightage: 30%

Assessment Tool Weightage: 40%

1. Identifying and prioritizing vulnerabilities

After discovering multiple open ports, I would first identify the services and their versions running on those ports. Then I would prioritize vulnerabilities based on their severity, exploitability, exposure to the Internet, and business impact. Critical services with known high-severity vulnerabilities would be addressed first, followed by medium- and low-risk issues.

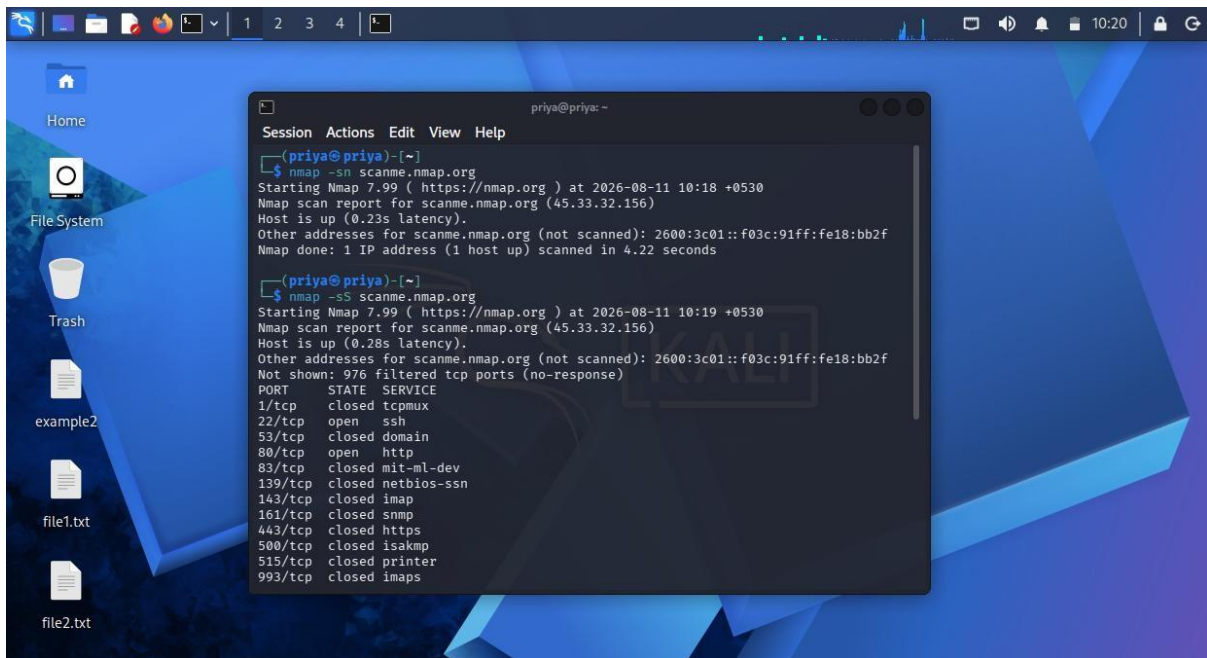
Step 1: Host discovery

Bash nmap -sn

scanme.nmap.org

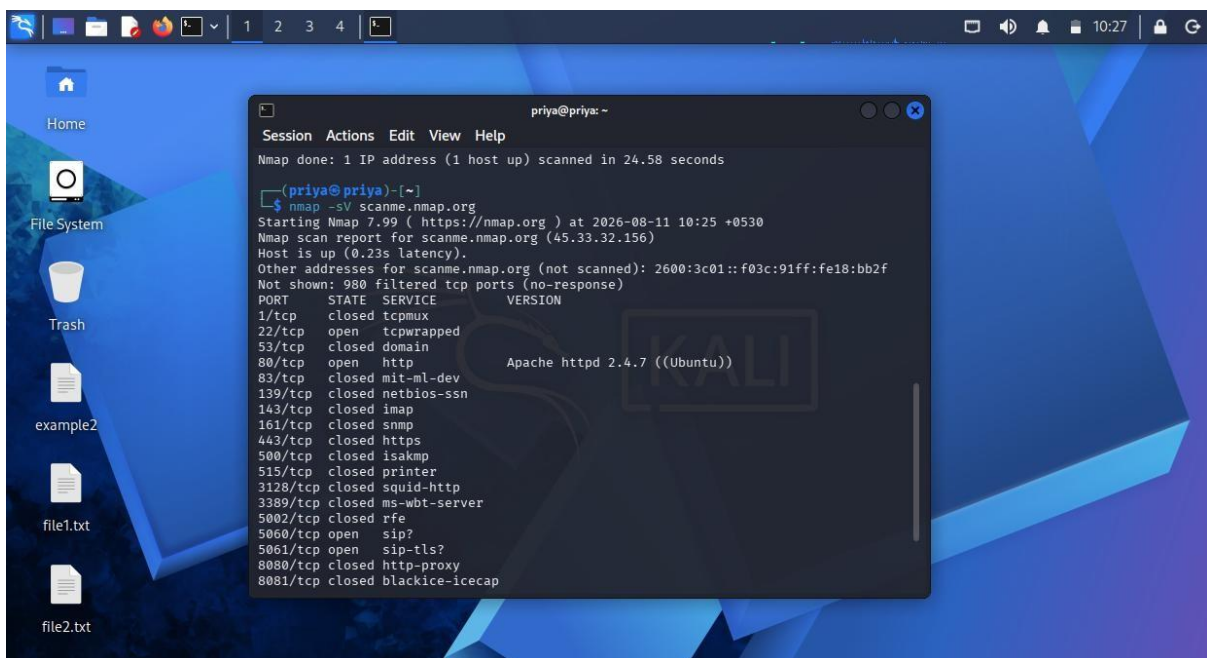
Step 2: TCP port scanning Bash

nmap -sS scanme.nmap.org



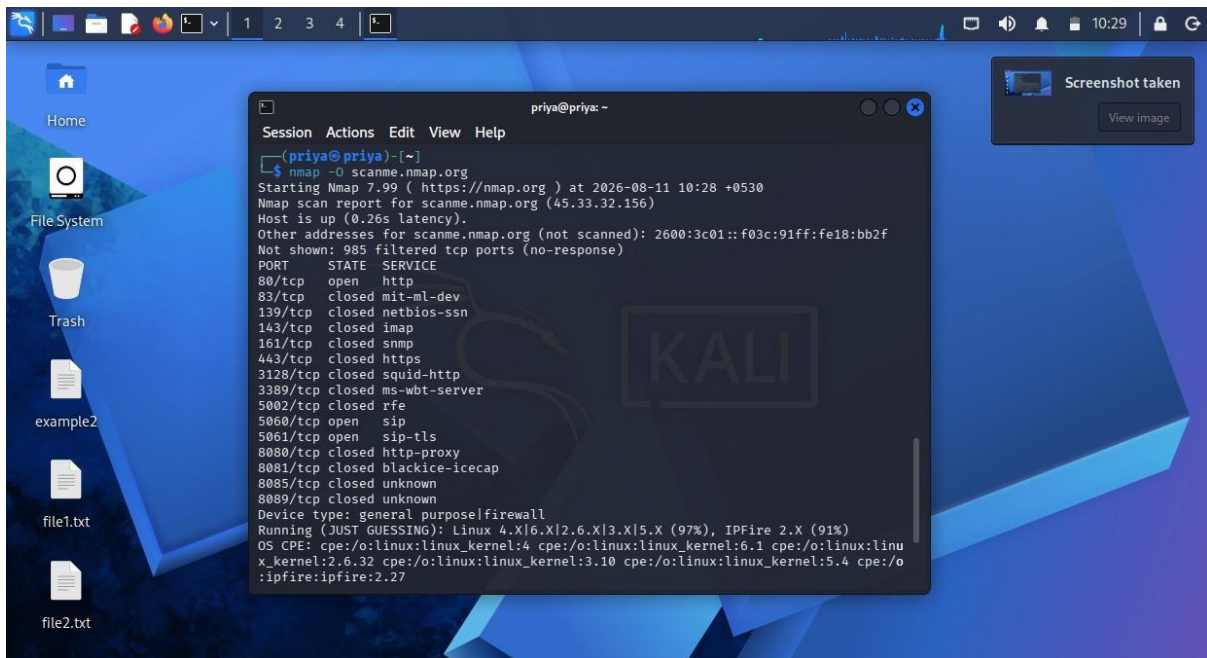
Step 3: Service and version detection

Bash `nmap -sV scanme.nmap.org`



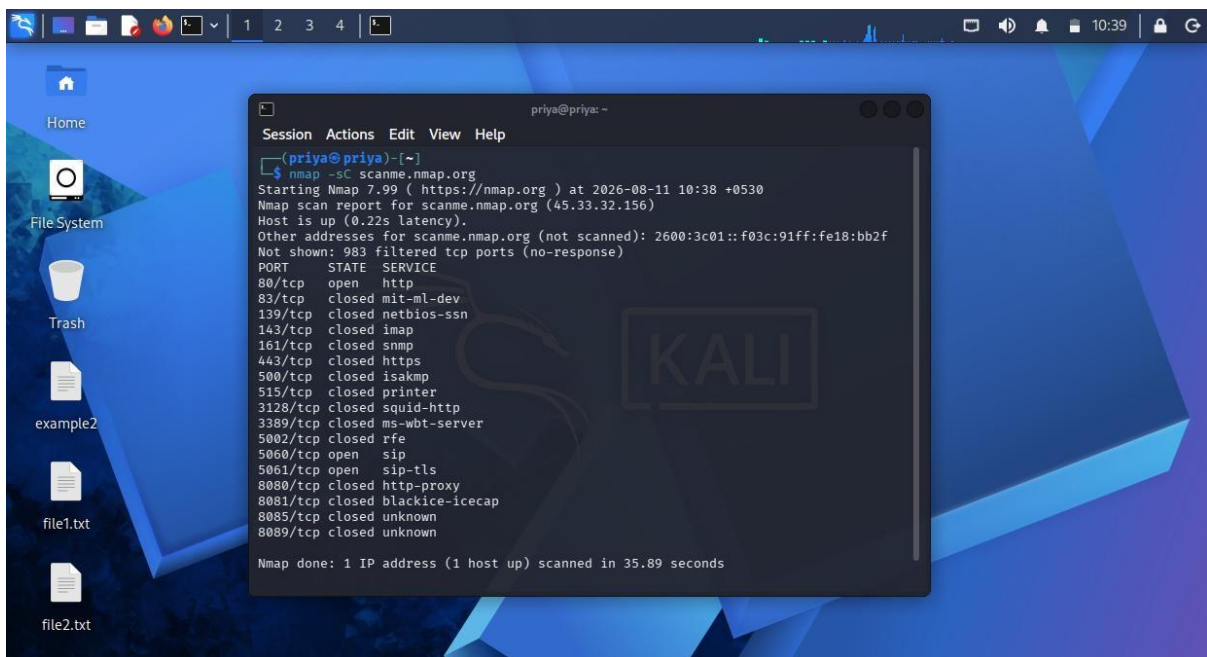
Step 4: Operating system detection Bash

`sudo nmap -O scanme.nmap.org`



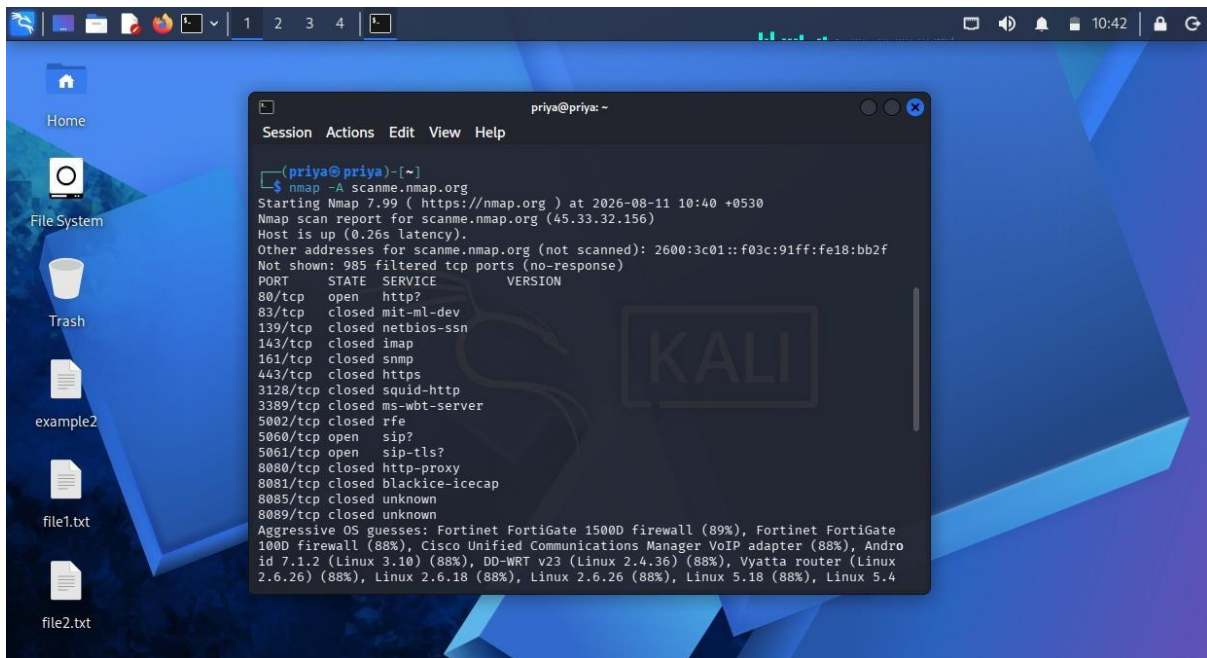
Step 5: Default script scanning Bash

`nmap -sC scanme.nmap.org`



Step 6: Complete enumeration Bash

`sudo nmap -A scanme.nmap.org`



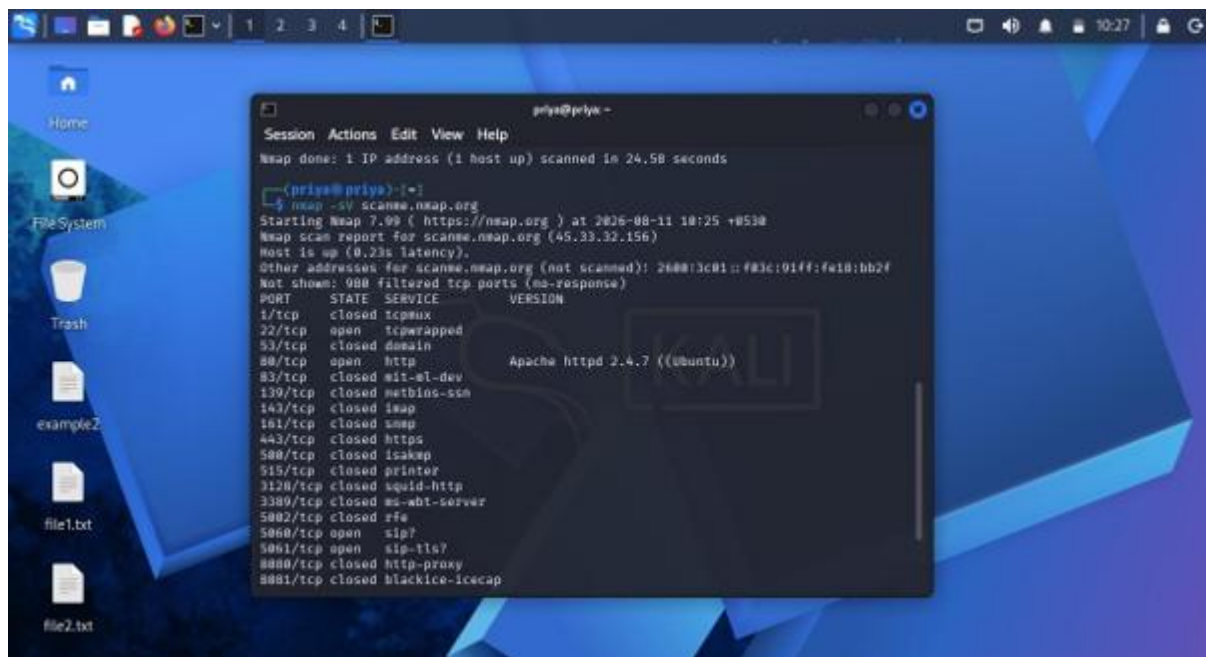
2 . Difference between SYN scan and TCP Connect scan

A SYN scan and TCP Connect scan may produce different results because of **firewalls, IDS/IPS systems, packet filtering, network latency, or different TCP connection handling**. A SYN scan does not normally complete the TCP connection, whereas a TCP Connect scan completes the connection. I would verify the findings by repeating both scans and comparing the results.

Command:

Bash

```
nmap -sS -sT <TARGET_IP>
```



3. Most ports appear

Step 1: Host discovery

Bash nmap -sn

192.168.56.102

Step 2: TCP port scan

Bash nmap -sS

192.168.56.102

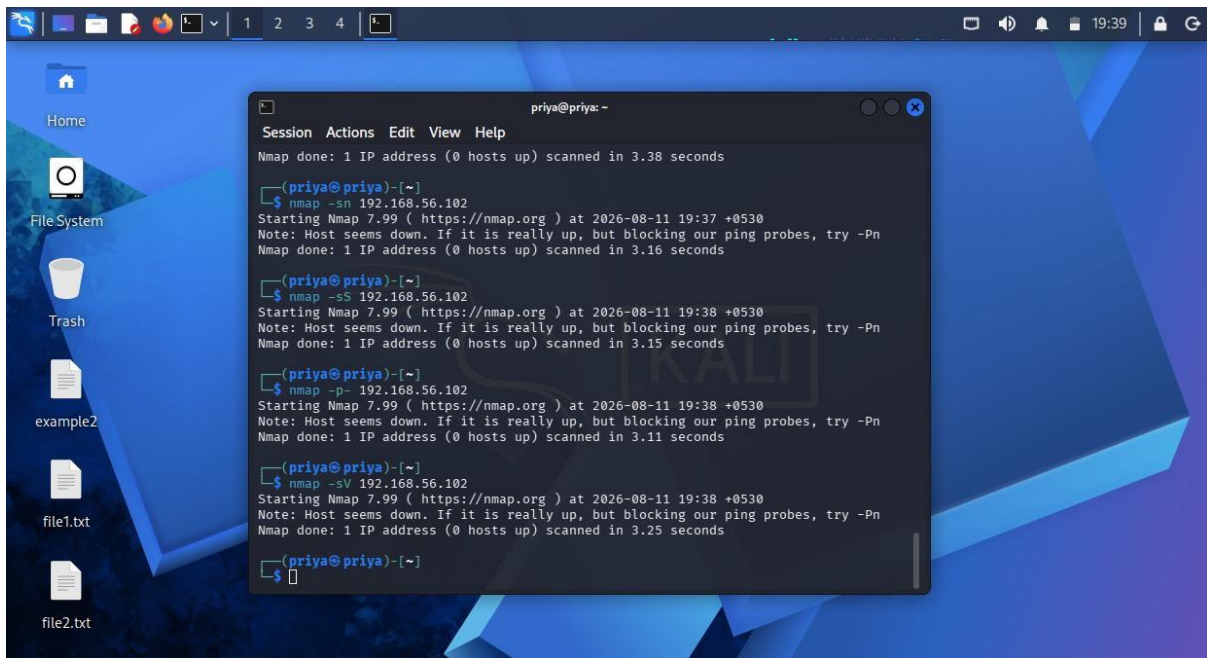
All TCP ports

Bash nmap -p- 192.168.56.102

Step 4: Service detection

Bash nmap -sV

192.168.56.102



Step 1: Host discovery

Bash `nmap -sn`

`192.168.56.103`

Step 2: Port scanning

Bash `nmap -sS`

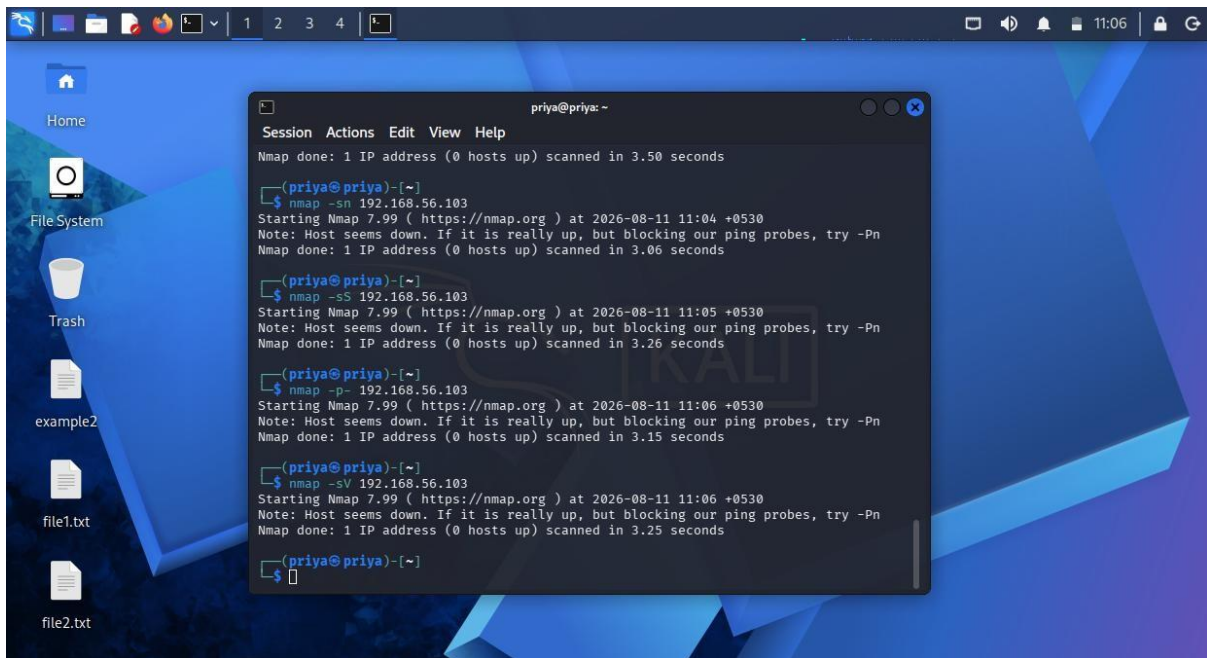
`192.168.56.103`

Step 3: All TCP ports

Bash `nmap -p- 192.168.56.103`

Step 4: Service/version detection

Bash `nmap -sV 192.168.56.103`



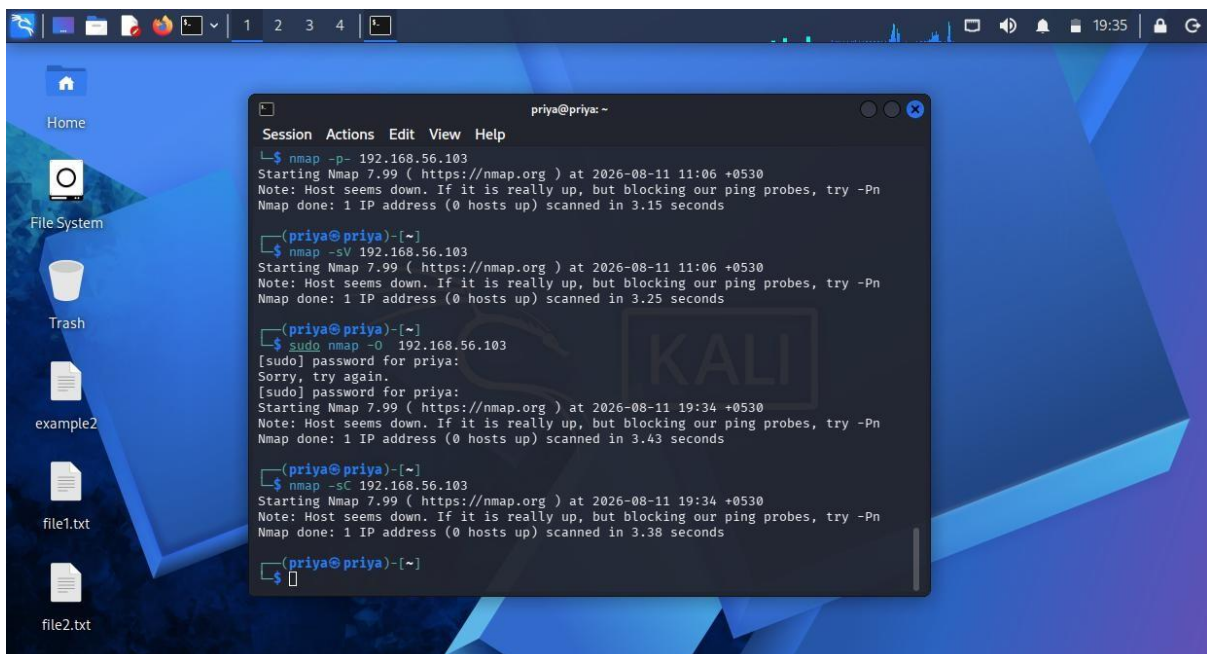
Step 5: OS detection Bash

`sudo nmap -O 192.168.56.103`

Step 6: Default NSE scripts

Bash `nmap -sC`

`192.168.56.103`



5. Assessing exposed SSH, FTP, and database services

Step 1: Host discovery

Bash nmap -sn

192.168.56.1

Step 2: TCP port scan

Bash nmap -sS

192.168.56.1

Step 3: Scan all TCP ports

Bash sudo nmap -p-

192.168.56.1

Step 4: Service detection

Bash nmap -sV

192.168.56.1

